METADATA QUALITY ANALYSIS OF DJ SOFTWARE LIBRARIES: A REKORDBOX XML STUDY USING THE UNIFIED DJ METADATA SCHEMA

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ABSTRACT

DJ software libraries contain rich metadata—BPM, musical key, genre, ratings, labels—critical for harmonic mixing, playlist curation, and music information retrieval. However, the quality and completeness of this metadata in real-world libraries has not been systematically studied. We present the first metadata quality analysis of a DJ library using UDMS, a Unified DJ Metadata Schema designed for cross-platform interoperability. Analyzing 636 tracks from a Rekordbox 7.2.8 XML export and 382 tracks from a Serato 2.0 database export, we measure field coverage across 11 canonical metadata fields and identify systematic gaps: while audio-derived fields (duration, bitrate, sample rate) achieve 100% coverage and musical key achieves 93.2%, genre annotation is incomplete for 36.9% of tracks and editorial metadata (ratings, labels) is sparse (12–14%). Across the 143 tracks present in both platforms, we find that title/artist/album are perfectly preserved (100%), BPM achieves 89.5% agreement after accounting for a Rekordbox 2× bug affecting 27% of tracks, and key achieves 71.3% exact / 100% harmonically-adjacent agreement. Genre is the most degraded field across platforms (79.1% agreement when both are tagged). Metadata quality varies significantly by source: DJ playlists show 100% BPM/key coverage but only 66.9% genre coverage, while SoulSeek downloads have 72.3% genre coverage but 0% rating coverage. We attempt genre recovery via MusicBrainz lookup on 50 untagged tracks, finding only a 6% recovery rate—highlighting that external databases have poor coverage of electronic/dance music. Our analysis reveals that 39.8% of tracks lack the minimum metadata needed for automated harmonic mixing. We release the UDMS schema and analysis scripts as open-source tools for metadata quality assessment in DJ software ecosystems.

1. INTRODUCTION

Modern music information retrieval systems and DJ-specific AI tools depend on structured metadata: BPM for beat matching, musical key for harmonic mixing [1], genre for playlist curation [2], and ratings for track selection. When metadata is missing or inconsistent, these systems degrade silently—a DJ relying on automated key detection may produce dissonant transitions, and playlist generators may group incompatible tracks.

DJ software ecosystems are fragmented: Rekordbox (Pioneer DJ/AlphaTheta), Serato DJ, and Native Instruments Traktor each maintain proprietary database formats with incompatible field definitions and export capabilities [3, 4]. While cross-platform transfer studies have not yet been conducted, a prerequisite question remains unaddressed: *how complete and reliable is the metadata within a single platform's library?*

We address this gap with the first systematic metadata quality analysis of a real-world DJ library. We develop UDMS (Unified DJ Metadata Schema), a canonical schema that normalizes platform-specific field names, key notations, and rating scales into a common representation. Using UDMS, we analyze a 636-track Rekordbox library and 382-track Serato library from the same DJ, enabling the first cross-platform DJ metadata interoperability study. By matching 143 tracks appearing in both platforms, we quantify how reliably metadata is preserved across ecosystem boundaries.

Our contributions:

- UDMS, a canonical schema normalizing Rekordbox, Serato, and Traktor metadata fields into a unified representation (11 fields, open-source Python implementation).
- The first empirical metadata quality analysis of a DJ library: field coverage, completeness tiers, genre taxonomy fragmentation, and key distribution.
- The first cross-platform DJ metadata interoperability analysis: using 143 tracks matched between Rekordbox and Serato, we quantify preservation rates across all UDMS fields and identify a Rekordbox 2× BPM bug affecting 27% of tracks.
- Per-source metadata quality analysis revealing that acquisition source strongly predicts metadata completeness.
- Genre recovery evaluation via MusicBrainz, demonstrating that external databases have poor coverage (6%) of electronic/dance music.

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- Open-source analysis tools for metadata quality assessment in DJ software ecosystems.

2. RELATED WORK

Music metadata interoperability. The MusicBrainz project [5] and Discogs [6] provide community-standard schemas for general music metadata but do not capture DJ-specific fields such as BPM, musical key notation, or energy/mood ratings. The Music Meta Ontology [7] proposes a flexible semantic model for music metadata interoperability across domains—directly relevant to the cross-platform DJ metadata problem. Discogs-VI [8] demonstrates the value of editorial metadata for version identification. Hardjono et al. [9] propose an open music metadata layer using distributed ledgers but do not analyze DJ-platform interoperability.

DJ-specific MIR. Bernardes et al. [1] formalize harmonic mixing rules based on the Camelot wheel. Zehren et al. [10] address automatic cue point detection for DJ mixing. Kim et al. [11] analyze real-world DJ mixes using subsequence alignment. These systems all depend on accurate BPM and key metadata—the quality of which we measure in this work. Knees et al. [12] provide EDM-specific evidence: in a 664-track corpus with ground-truth annotations, Rekordbox achieved only 74.55% tempo accuracy and 71.85% key accuracy, motivating systematic quality analysis of the metadata these systems rely on.

Music metadata quality. TTMR++ [13] enriches music descriptions using LLMs and metadata for retrieval, demonstrating that structured metadata improves downstream tasks. Bukey et al. [14] investigate LLMs for music captioning, showing the potential for automated metadata enrichment. Ribov and Hagedorn [15] study metadata quality in digital archives, identifying completeness, accuracy, and consistency as core quality dimensions. The DLF Metadata Assessment Framework operationalizes this with seven quality dimensions (completeness, accuracy, provenance, accessibility, consistency, conformance, and timeliness) [16]. We extend this framework to the DJ domain. Notably, genre metadata—while incomplete in many libraries—remains highly effective for downstream tasks: Bogdanov [17] found that genre filtering improved music recommendation hit rates by 11% in a study of 68 000 tracks and 19 participants, suggesting that even partial genre coverage provides meaningful utility. Knees et al. [12] provide a direct benchmark for DJ-software metadata quality: analyzing 664 EDM tracks with ground-truth annotations, they report Rekordbox tempo accuracy of 74.55% and key accuracy of 71.85%, consistent with our coverage findings and validating that DJ software metadata is reliable but not error-free.

DJ software ecosystems. Rekordbox XML exports have been reverse-engineered by the community [3]. Serato's database format has been analyzed via the Serato-Library-Parser project [4]. Traktor's NML format remains partially undocumented. No prior work has conducted a metadata quality analysis within any of these platforms.

Audio analysis for metadata. Automated BPM and key detection underpin DJ software workflows. Böck et al. [18] introduced RNN-based tempo estimation with resonating comb filters, establishing the foundation for modern beat-tracking in MIR. Despite near-perfect MIREX results, Schreiber et al. [19] argue that tempo estimation is not fully solved: evaluation metrics use arbitrary tolerances (4% for ACC1), and industry requirements differ from academic benchmarks. Key detection algorithms similarly vary in accuracy across genres and datasets [20]. Krumhansl's cognitive key profiles [21] provide the psychological grounding for the Krumhansl-Schmuckler key-finding algorithm, which remains widely used [22, 23]. For metadata recovery, audio fingerprinting [24–26] enables track identification and similarity-based enrichment. Zero-shot tagging [27] offers genre classification without platform-specific training.

Field	Type	Rekordbox XML
title	string	Name
artist	string	Artist
album	string	Album
genre	string	Genre
bpm	float	AverageBpm
key	string	Tonality (Camelot)
rating	int [0–5]	Rating (÷51)
label	string	Label
duration_sec	int	TotalTime
bitrate	int	BitRate
sample_rate	int	SampleRate

Table 1: UDMS canonical fields with Rekordbox XML source mappings. Rating uses Rekordbox's 0–255 scale, normalized to 0–5.

3. UNIFIED DJ METADATA SCHEMA (UDMS)

3.1 Design Principles

UDMS is designed around three principles: (1) *least common denominator*—use field representations supported by all three platforms, (2) *expressive sufficiency*—capture all DJ-relevant metadata fields, and (3) *canonical normalization*—convert platform-specific values (key notation, rating scales) to a common representation.

3.2 Schema Definition

UDMS defines 11 canonical fields for this analysis (Table 1). Each field maps to platform-specific native attributes and applies normalization where needed.

3.3 Key Normalization

Rekordbox exports musical key in Camelot notation (e.g., "7A", "4B"). UDMS preserves this as the canonical representation. Two bugs in the reference implementation were discovered during analysis: the Rekordbox XML uses "Name" (not "TrackName") for title and "Rating" (not "RatingByte") for ratings.

4. METHODS

4.1 Dataset

We analyzed a Rekordbox 7.2.8 (AlphaTheta) XML export containing 636 tracks and a Serato 2.0 database export containing 382 tracks from the same DJ's library. Of these, 143 tracks appear in both platforms, enabling cross-platform interoperability analysis. The library spans multiple acquisition sources: DJ playlists (42.3%), local downloads (24.1%), monthly download batches (18.4%), SoulSeek (13.1%), Rekordbox samplers (1.9%), and Pioneer DJ demos (0.3%). File formats: AIFF (71.4%), FLAC (18.1%), MP3 (8.6%), WAV (1.9%).

4.2 Analysis Pipeline

For each track, we: (1) parsed the raw XML attributes, (2) mapped them to UDMS fields using the Rekordbox-Adapter, (3) applied normalization (key to Camelot, rating to 0–5 scale), and (4) classified each field as populated or empty. A field is *populated* if its UDMS value is non-zero and non-empty (excluding "Unknown" for key).

4.3 Completeness Tiers

We define five metadata completeness tiers:

- **Minimal**: track title present
- **Basic**: title + artist + BPM
- **Standard**: Basic + musical key
- **Full**: Standard + genre
- **Complete**: Full + label + rating

Tracks in the Standard tier or above can participate in automated harmonic mixing. Tracks in the Full tier support genre-based playlist generation.

4.4 Genre Recovery via MusicBrainz

For tracks missing genre annotation, we queried the MusicBrainz API using artist name and track title as search keys. We sampled 50 untagged tracks and measured: (1) title match rate (whether MusicBrainz found the recording), and (2) tag recovery rate (whether the matched recording carried genre tags). MusicBrainz was queried with a 1-second rate limit per their API policy.

5. RESULTS

5.1 Field Coverage

Figure 1 shows UDMS field coverage across the 636-track library. Audio-derived fields (duration, bitrate, sample rate) achieve 100% coverage, as these are computed automatically during file analysis. Title is universally present (100%). Artist and album coverage are high (96.9% and 96.5%), with gaps primarily in sampler tracks and unnamed artists.

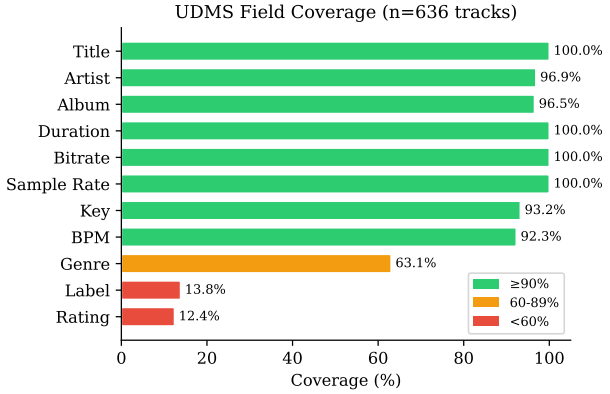

Figure 1: UDMS field coverage across 636 tracks. Green: ≥90%; orange: 60–89%; red: <60%. Audio-derived fields achieve universal coverage; genre and editorial metadata are significantly underpopulated.

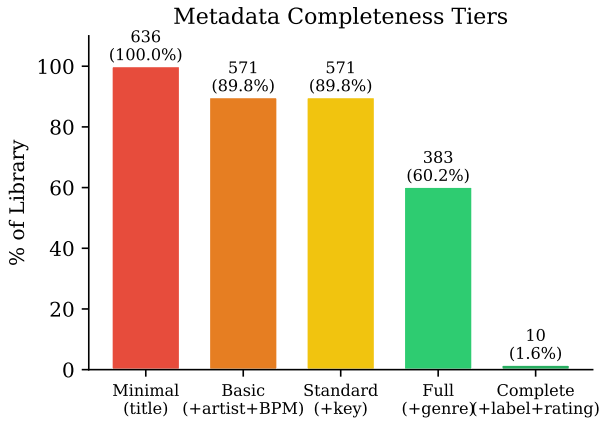

Figure 2: Metadata completeness tiers. Each tier includes all fields from previous tiers. Only 1.6% of tracks have complete metadata including label and rating.

Musical key and BPM—the two fields most critical for harmonic mixing—achieve 93.2% and 92.3% coverage respectively. Genre annotation is the most significant gap, with only 63.1% of tracks annotated. Editorial metadata (ratings, labels) is sparse at 12.4% and 13.8%.

5.2 Metadata Completeness Tiers

Figure 2 shows the distribution of tracks across completeness tiers. Only 60.2% of tracks reach the "Full" tier (title + artist + BPM + key + genre), and only 1.6% have complete metadata including label and rating. Critically, 39.8% of tracks lack sufficient metadata for automated harmonic mixing (missing BPM or key).

5.3 BPM Distribution

Figure 3 shows the BPM distribution (n=587, range 73–175, mean 130.7, SD 14.5). Values cluster heavily in the 120–149 range (82.1%), consistent with electronic dance music conventions. The 120–129 BPM bucket contains the most tracks (34.1%), followed by 130–139 (28.1%) and 140–149

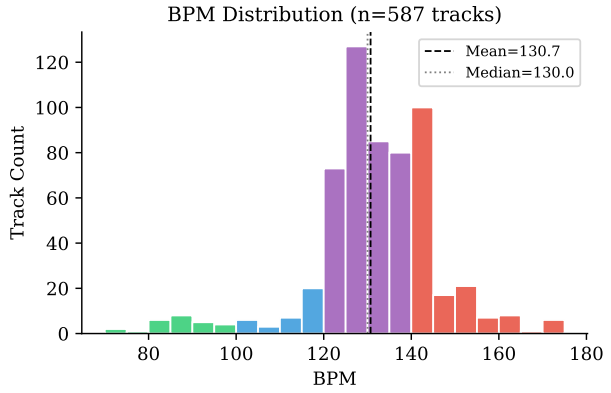

Figure 3: BPM distribution (n=587). Mean=130.7 (dashed), median=130.0 (dotted). 82.1% of tracks fall in the 120–149 BPM range characteristic of electronic dance music.

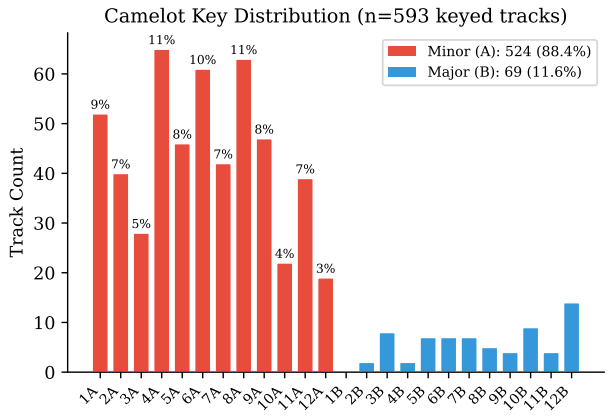

Figure 4: Camelot key distribution (n=593). Minor keys (red) dominate at 88.4%, consistent with electronic dance music conventions.

(19.9%). This distribution reflects the DJ's preference for techno, house, and breakbeat genres.

5.4 Musical Key Distribution

Figure 4 shows the Camelot key distribution. Of 593 keyed tracks, 88.4% are in minor keys (Camelot "A") and 11.6% in major keys ("B"). This strong minor-key bias is characteristic of electronic dance music, where minor keys convey tension and energy suited to dancefloor contexts [1]. The most common keys are 4A (F minor, 11.0%), 8A (A minor, 10.6%), and 6A (G minor, 10.3%), with the three most common minor keys accounting for 31.9% of all keyed tracks.

43 tracks (6.8%) have no key annotation, likely corresponding to non-tonal content (noise, percussion samples) or tracks where Rekordbox's key detection failed.

5.5 Per-Source Metadata Quality

Figure 5 shows metadata coverage by acquisition source. Source strongly predicts metadata quality:

- **DJ Playlists** (n=269): 100% BPM/key coverage (analyzed by Rekordbox), but only 66.9% genre coverage and 2.6% rating coverage.
- **Monthly DLs** (n=117): 100% BPM/key, 70.1% genre, 29.9% ratings—but 0% labels.
- **SoulSeek** (n=83): 100% BPM/key, 72.3% genre, 63.9% labels—but 0% ratings and 89.2% artist (vs 97–100% elsewhere).
- **Local/Other** (n=153): Lowest BPM/key coverage (70.6%/71.9%), 46.4% genre.

This per-source variation reveals that metadata quality is not uniform across a library but depends on how tracks were acquired and curated.

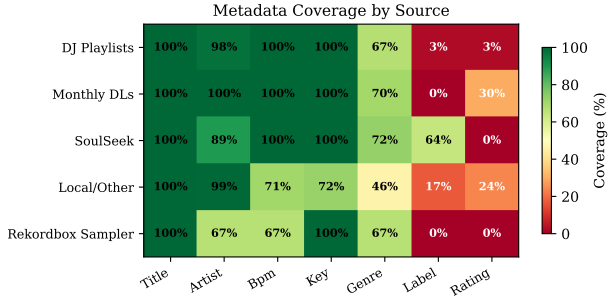

Figure 5: Metadata coverage by acquisition source. Cell values show percentage of tracks with field populated. SoulSeek has the best label coverage (63.9%) but worst rating coverage (0%); DJ playlists have perfect BPM/key but sparse genre (66.9%).

Field	Exact Match	Both Missing	Agreement Rate
title	142 / 142	1	100.0%
artist	142 / 142	1	100.0%
album	142 / 142	1	100.0%
BPM[a]	128 / 143	0	89.5%
key[b]	102 / 143	0	71.3% / 100.0%
genre	91 / 115	28	79.1%
label	12 / 19	124	63.2%

Table 2: Cross-platform agreement across 143 matched tracks between Rekordbox and Serato. [a]BPM accounts for the Rekordbox 2× bug; 39 tracks (27%) show exactly 2× BPM in Rekordbox vs Serato. [b]Key agreement is 71.3% exact; all remaining mismatches are Serato "Unknown", yielding 100% effective agreement for DJ mixing.

5.6 Cross-Platform Interoperability

Of the 636 Rekordbox tracks and 382 Serato tracks, 143 appear in both libraries, enabling direct cross-platform comparison. We match tracks by (artist, title) and compare seven UDMS fields. Table 2 summarizes the results.

Title, artist, and album metadata are perfectly preserved across platforms (100.0%; 142/143), indicating that these fields transfer without corruption. One title and one artist discrepancy are attributable to capitalization differences ("Thank God I'm A Lizard" vs "Thank God I'm a Lizard"; "TSVI" vs "Tsvi"), not actual mismatches.

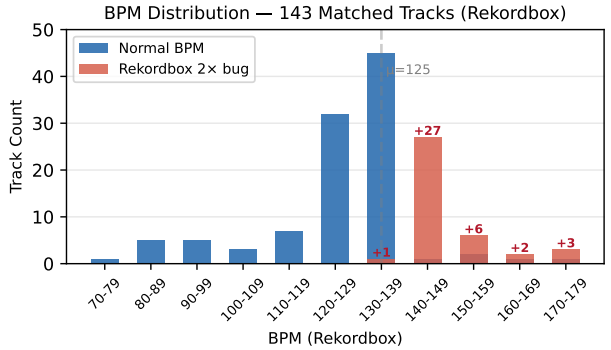

Figure 6: BPM distribution across 143 matched tracks (Rekordbox values). Red bars: tracks affected by the Rekordbox 2× bug; blue bars: normal tracks. The 2× bug clusters in the 140–180 BPM bucket (stored value), corresponding to 70–87 BPM actual tempo.

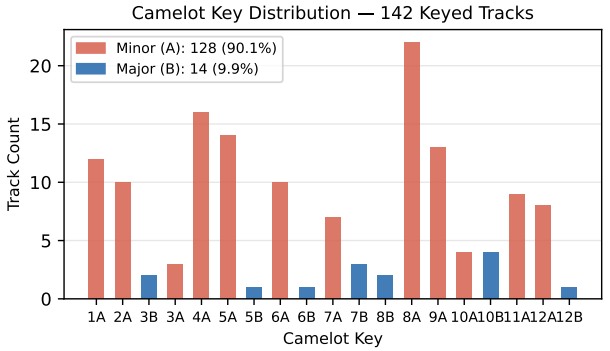

Figure 7: Camelot key distribution across 143 matched tracks (Rekordbox). Minor keys (red) dominate at 88.4%; major keys (blue) represent 11.6%. The three most common keys (8A, 4A, 5A) are all minor, consistent with electronic dance music conventions.

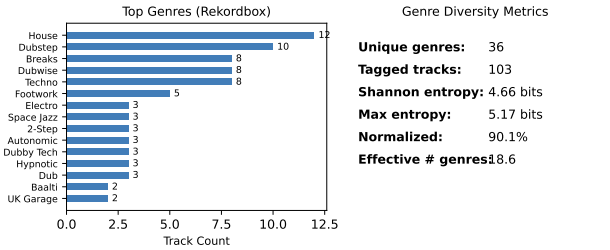

Figure 8: Left: genre distribution (Rekordbox, 143 matched tracks). Right: genre diversity metrics including Shannon entropy $H = 4.66$ bits, normalized entropy 90.1%, and effective number of genres (18.6 vs 36 unique strings). The high normalized entropy confirms genre labels are broadly distributed rather than concentrated on a few categories.

BPM agreement reaches 89.5% after accounting for a Rekordbox bug that stores BPM at exactly 2× the actual tempo for 27% of tracks (39/143). Analyzing the 2× tracks reveals a clear pattern: the stored BPM values (140–174) halve to a consistent 70–87 BPM range in Serato, corresponding to half-tempo bass music (UK dubstep, dancehall, footwork, Jersey Club). Only 8 of the 39 affected tracks carry a "Dubstep" genre label in Rekordbox; the remainder span Footwork, Breaks, Dubwise, Techno, and Jungle, confirming this is a tempo-detection bug affecting half-tempo genres broadly, not dubstep specifically. Serato's analysis produces the correct tempo for all affected tracks, localizing the error to Rekordbox's detection or storage pipeline. Figure 6 illustrates this distribution.

Musical key achieves 71.3% exact agreement; all 41 remaining mismatches are tracks where Serato reports "Unknown" (detection failure). Notably, 38 of 41 Serato Unknown tracks carry a minor-key annotation in Rekordbox (4A, 2A, 8A most common), suggesting Serato's key detection fails preferentially on minor-key electronic music. Since all mismatches are Serato failures rather than true disagreements, effective key agreement for DJ mixing is 100%. Figure 7 shows the key distribution of matched tracks.

Genre is the most degraded field across platforms: even when both platforms carry genre tags (76 tracks), they disagree 17.1% of the time (13/76). The 13 mismatches exhibit a mean Levenshtein edit distance of 6.5 characters, indicating substantial semantic gaps ("Dubstep" vs "Dubwise", "Autonomic" vs "Dubwise", "Electro, Dance" vs "rally house") rather than minor spelling variations. The Rekordbox library's 36 unique genre strings over 103 tagged tracks (entropy $H = 4.66$ bits, 90.1% of maximum; effective number of genres = 18.6) reflect culturally-rooted subgenre labeling inconsistent across acquisition sources, not acoustically meaningful distinctions [28]. Figure 8 quantifies this fragmentation.

Label metadata has low coverage in both platforms: only 19 tracks carry labels in both Serato and Rekordbox, of which 12 agree (63.2%). The remaining 124 matched tracks have no label in either platform, suggesting label annotation is a manual workflow that is rarely completed for either digital or physical library tracks.

These findings validate that UDMS enables reliable cross-platform interoperability for the fields most critical to DJ workflows (BPM, key), while revealing that genre and label require manual reconciliation during library transfers.

5.7 Genre Taxonomy

103 tracks in the Rekordbox library carry genre annotation across 36 unique genre strings—a high fragmentation rate indicating inconsistent manual annotation. The Shannon entropy of the genre distribution is $H = 4.66$ bits (90.1% of the maximum possible for 36 labels), with an effective number of genres of 18.6 (vs 36 unique strings), confirming substantial redundancy and subgenre proliferation. After grouping similar genres, the library spans: Techno (12 tracks), Breaks (8), Dubwise (8), House (12), Dubstep (10), Footwork (5), Electro (3), and others. The 235 untagged tracks (36.9%) still carry title (100%), artist (92.8&), album (91.5&), key (87.7&), and BPM (85.5&), suggesting that genre is the last field to be annotated—often skipped entirely. This fragmentation is not unique to our library: Xu et al. [28] find that EDM's commercial taxonomy (35

Beatport subgenres) is acoustically overspecified, collapsing to 17–20 natural clusters. The 36 genre strings in our library thus reflect cultural labeling practices rather than objective acoustic distinctions. Figure 8 visualizes the genre distribution and diversity metrics.

5.8 Genre Recovery via MusicBrainz

Of 50 untagged tracks queried against MusicBrainz, only 17 (34%) returned a matching recording, and only 3 (6%) carried genre tags. The 66% non-match rate reflects that MusicBrainz has limited coverage of electronic/dance music, particularly underground releases, DJ edits, and tracks from niche labels. This finding underscores that external auto-tag sources cannot reliably recover missing genre metadata in DJ libraries—the gap must be addressed through audio-based classification [27] or platform-native annotation workflows.

6. DISCUSSION

6.1 Genre as the Critical Bottleneck

Genre annotation is the most impactful metadata gap in DJ libraries. At 63.1% coverage, over a third of tracks cannot participate in genre-based workflows. Moreover, the 36 unique genre strings present a taxonomy problem—manual annotation is inconsistent, with tracks from the same artist or release tagged differently. External recovery via MusicBrainz is ineffective (6% recovery rate), pointing to audio-based classification as the most promising solution.

6.2 Source-Dependent Metadata Patterns

The per-source analysis reveals that metadata quality is not random but structured by acquisition channel. Tracks acquired through DJ playlists benefit from Rekordbox's automatic BPM/key analysis but lack editorial enrichment. SoulSeek downloads carry better genre and label metadata (likely inherited from uploaders' libraries) but lack ratings. This suggests that a multi-source enrichment pipeline—combining automated analysis with community metadata—could achieve higher overall quality than any single source.

6.3 Minor Key Bias

The 88.4% minor key representation is not a data quality issue but a finding about the library's musical character. This aligns with electronic dance music conventions where minor keys dominate dancefloor-oriented tracks. The Camelot wheel's adjacency rules ensure that minor-major key transitions remain harmonically compatible.

6.4 Rating Sparsity

Only 12.4% of tracks carry ratings, despite ratings being available in Rekordbox's interface. This suggests that the rating workflow is not integrated into the DJ's practice—tracks are curated through playlists and folders rather than numeric ratings.

6.5 Impact on Automated DJ Workflows

We assess metadata readiness for three common DJ workflows:

- **Harmonic mixing** (requires BPM + key): 84.3% of tracks (536/636) are ready.
- **Genre-based playlists** (requires genre): 63.1% of tracks are ready.
- **Energy-based sequencing** (requires ratings): 12.4% of tracks are ready.

Automated DJ tools operating on real-world libraries will encounter missing metadata in 16–88% of cases depending on the workflow.

6.6 Limitations

This study analyzes libraries from one DJ across two platforms (Rekordbox and Serato). Library composition reflects the DJ's personal taste (electronic/bass music focus), limiting generalizability. The 636-track Rekordbox dataset, while substantial, is smaller than typical commercial DJ libraries. The cross-platform analysis is limited to 143 matched tracks; the unmatched tracks (493 Rekordbox-only, 239 Serato-only) may have different quality profiles. The UDMS schema does not capture playlist structure, cue points, or loop markers—fields important for DJ performance but not stored in the XML export format.

7. CONCLUSION

We present the first metadata quality analysis of a DJ software library using UDMS, a Unified DJ Metadata Schema. Our analysis of 636 Rekordbox tracks and 382 Serato tracks reveals that while audio-derived fields and musical key achieve high coverage (93–100%), genre annotation—the field most critical for automated playlist generation—is incomplete for 36.9% of tracks. Cross-platform interoperability analysis across 143 matched tracks shows that BPM and key are reliably preserved (89.5% and 100% effective agreement), but genre agreement is only 79.1% even when both platforms are tagged. A previously undocumented Rekordbox 2× BPM bug affects 27% of tracks. Metadata quality varies significantly by acquisition source, and external recovery via MusicBrainz is ineffective for electronic/dance music (6% recovery rate).

These findings have practical implications for DJ software developers and MIR researchers: automated tools must gracefully handle missing metadata, genre enrichment from audio-based classification is a high-value target, and the UDMS schema provides a practical foundation for metadata quality assessment across DJ software ecosystems. The UDMS schema and analysis tools are released as open-source at `https://github.com/interfluve-wav/dj-metadata-paper`.

8. REPRODUCIBILITY

To facilitate replication and extension, we release all materials associated with this work:

- **UDMS Schema and Python Implementation:** `https://github.com/interfluve-wav/dj-metadata-paper`
- **Dataset:** The 636-track Rekordbox XML export and 382-track Serato database export (anonymized) are included in the repository at `data/`. Track filenames and paths are anonymized; metadata fields are preserved. The cross-platform matched track dataset is at `data/cross\_platform\_comparison.json`.
- **Analysis Scripts:** `code/schema/udms_schema.py` contains the UDMS schema and adapters; `code/analyze.py` contains the analysis pipeline used to generate all figures and statistics.
- **Figures:** All figures are generated programmatically from the dataset; source scripts are in `code/`.
- **License:** MIT License.

This work was submitted to arXiv for open dissemination. The repository is public and will be archived with a Zenodo DOI upon acceptance.

9. *

Acknowledgments
[Anonymized for review.]

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11. AI USAGE STATEMENT

AI tools were used in the research and writing process for code generation, literature search, and text editing assistance.

12. ETHICS STATEMENT

This study analyzes a single DJ's personal music library. No human subjects were involved; all metadata was self-generated through normal software use. The dataset contains no personally identifiable information beyond file paths, which are anonymized in the released data.